

From 3D Semantic Segmentation to Qualitative Neural Rendering:

Object Boundaries, Multisensory Registration, and Tests of Differentiated Content

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August 13, 2026

Abstract

A semantic-segmentation system can label every visible point without explaining why a biological observer experiences one bounded object rather than an inventory of classified features. The missing problem is operational. Local features must be assigned to objects and surfaces; borders must acquire ownership; partial observations must be completed without erasing uncertainty; visual, auditory, tactile, proprioceptive, vestibular, interoceptive, memory, and action signals must be transformed among reference frames; and the resulting hypotheses must alter recognition, report, choice, movement, and subsequent learning. This paper develops a source-faithful Self Aware Networks (SAN) bridge from segmentation to **qualitative neural rendering**. The proposal treats qualitative content as a distributed, receiver-relative, recurrently consequential state - not as a picture inspected by a hidden observer and not as a property conferred by a class label.

The model separates semantic class, instance identity, border ownership, spatial registration, tonic receiver state, Phase Wave Differential (PWD), recurrent object hypothesis, and action-return error. Fifty equations specify those variables, their domains, normalized objectives, estimator choices, causal assumptions, model-separation tests, sample-budget logic, calibration, distribution shift, and local stability conditions, preventing one evidentiary level from standing in for another. Six hypotheses and an Object-Boundary Registration Benchmark compare a feedforward feature model, a semantic-only model, an instance-and-border model, a recurrent-completion model, a multisensory-registration model, the full receiver-state/PWD model, and capacity-matched global and latent alternatives. The executable reference application and trainable synthetic pilot establish executable and fitted synthetic development gates. The distribution-shift and latent-baseline stress test then makes the comparison materially harder: four 13-feature, 112-parameter representations are fitted under IID, correlation-reversal, noise, missingness, confound, and prevalence shifts, using five development and two independent replication seeds. Across 168 model fits and 126 typed-channel sham fits, M5's receiver-task mean remains above the global summary in every scenario, but the stronger unsupervised and supervised latent baselines equal or exceed M5 under development noise and missingness shifts. Missingness-shift replication preserves that reversal by approximately 0.024-0.025 accuracy. Fourteen development and four replication comparison or sham cells are nonpositive and remain in the record. Calibration error is substantial, particularly under confound reversal, so no probability-calibration or mechanism-identification claim is admitted. The result is a stress-tested synthetic development finding, not biological, phenomenal, or consciousness validation. The final test remains sealed. The paper also freezes a staged biological translation protocol separating stimulus, attention, report, eye movement, receiver state,

perturbation uptake, content-specific consequence, and rescue. Existing work on border ownership, recurrent completion, reference-frame transformation, sensory substitution, cortical stimulation, and perceptography supplies the measurable components. SAN’s contribution is the joined operator and the experimental demand that differentiated content be tracked from object organization through receiver transformation to causal network consequence.

The visual evidence and calibration readback adds a six-figure, table-derived evidence layer and an exact visual provenance ledger. The figures do not introduce another model or select a more favorable endpoint. They make the benchmark architecture, model parity, development and replication performance, comparator gaps, calibration defects, typed-channel sham behavior, and claim-promotion boundary inspectable in one readback. Every panel is generated from the canonical distribution-shift and latent-baseline stress-test tables or from a deterministic reproduction of the same training and validation routes. The visual audit preserves the result most likely to disappear in a promotional summary: latent alternatives outperform M5 under missingness, receiver-condition sham separation nearly collapses under confound reversal, and none of the bounded scores is adequately calibrated as a biological probability. A visual layer therefore counts as evidence organization, not evidentiary promotion.

1 Begin with the ordinary experience

Consider a plate containing steak and broccoli. The retinal image changes whenever the eyes move. A fork occludes part of the broccoli. Steam changes local contrast. The smell is not arranged on the retina, the texture reaches the fingers and mouth, and the sound of the fork arrives through another sensory pathway. Yet the broccoli is experienced as one object with a location, boundaries, properties, expected taste, and possible actions.

That stability is not given directly to the brain. Different receptors sample different variables at different locations and times. Their signals travel through pathways with different delays, coordinate systems, resolutions, and histories. The brain must determine which green edges belong to the broccoli, which belong to the plate, whether two visible fragments are one partly occluded object or two objects, and whether a smell, a remembered taste, and the pressure at the fork refer to the same thing. Movement then changes the evidence and tests the current answer.

Machine vision provides useful language for part of this task. Semantic segmentation assigns a class to each pixel or point. Instance segmentation separates two objects of the same class. Three-dimensional segmentation assigns labels to points, voxels, or surfaces. These operations answer versions of *what is where?* They do not, by themselves, answer four further questions:

1. Which side owns a shared border?
2. How are partial and cross-modal signals registered to the same object and body-world relation?
3. How does an arriving pattern change a receiver that is already active and historically tuned?
4. Which organized changes become causally available as this content rather than another?

The scientific problem is therefore not to rename segmentation as consciousness. It is to state the additional transformations between an object-wise representation and a differentiated neural state, and then to test those transformations independently.

2 What established science already provides

2.1 Labels, instances, and three-dimensional membership

Fully convolutional networks made dense per-pixel semantic prediction a standard task [1]. Mask R-CNN separated class membership from instance-specific masks [2]. PointNet and PointNet++ showed how unordered 3D point sets can support classification and segmentation, with the latter adding hierarchical local context across varying densities [3,4]. ScanNet joined reconstructed indoor geometry with semantic and instance annotations [5], while panoptic segmentation joined countable “things” and background “stuff” in one output format [6].

These distinctions matter because two pieces of broccoli can share a class while remaining different instances, and one partly hidden piece can occupy several disconnected image regions while remaining one object. A scientifically useful neural theory must keep class, instance, surface, border, location, and current visibility separate rather than compressing them into one word such as *representation*.

2.2 Biological borders are relational

The same local edge can belong to the figure on its left or the figure on its right. Macaque V2 neurons can signal this border-ownership relation even when local contrast in the receptive field is held similar [7]. Figure-ground modulation in V1 emerges later than the initial local visual response [8], is selectively reduced under anesthesia while basic receptive-field tuning remains [9], and can be pharmacologically dissociated from much of the feedforward response [10]. Columnar organization supplies additional structure to border-ownership signals [14].

This is already more than pixel labeling. Border ownership is relational: a boundary is represented together with the side or surface to which it belongs. It supplies a biological precursor for an object-wise constraint, while leaving open how that constraint is joined with sound, touch, body position, memory, action, and qualitative report.

2.3 Partial objects recruit additional computation

Occlusion exposes the limits of one-pass recognition. Humans can recognize severely partial objects, but masking disrupts this ability; human ventral responses to partial objects are delayed; and recurrently augmented models can recover performance that standard feedforward models lose [11]. Images with longer object-solution times also reveal performance not captured by standard feedforward models [12]. Critically, the need for recurrence varies with the problem: increasing scene-segmentation demand increases the cost of disrupting recurrent processing [13]. Broader recurrent-vision and object-representation syntheses supply the surrounding comparison class [21,22].

Recurrence therefore earns a specific role. It can revise, complete, suppress, or stabilize an object hypothesis when the first sweep is insufficient. The word *recurrent* is not yet an explanation of which relation is revised, which receiver changes, or which content follows. Those variables must remain explicit.

2.4 The brain transforms among reference frames

Multisensory registration does not require every signal to enter one fixed coordinate system. Visual and tactile responses in macaque ventral intraparietal cortex span mixtures from eye-centered to head-centered coordinates [16]. Human parietal cortex contains aligned near-face visual and tactile maps [17]. During tactile-motor transformation, parietal activity shifts from anatomical coding toward external spatial coding needed for action [18].

The useful principle is a family of transforms, not a universal screen. An object hypothesis can be registered relative to retina, eye, head, hand, body, reachable space, or world, depending on the receiver and task. A stable percept can arise from coordinated transformations among these frames while the underlying coordinates remain distributed.

2.5 Content depends on ongoing state

Conscious object-recognition studies separate low-level visibility, category report, and subjective recognition. Whole-brain 7 T fMRI found broad overlap between recognized and unrecognized trials, with recognition associated with response-magnitude differences and content decoding across distributed cortical networks [19]. Prestimulus ongoing activity can influence perceptual sensitivity, response criterion, and category decoding through partly different distributed networks [20].

An incoming image therefore does not act on a blank device. The same stimulus enters a receiver with a current excitability, inhibition, expectation, phase relation, attention state, body state, and recent history. The receiver’s state is part of the transformation. This observation supplies an empirical bridge to SAN’s tonic-context and receiver-relative formulation.

2.6 Intervention reveals network consequence

Modern stimulation studies demonstrate why encoding and causation must be separated. Dynamic sequences of visual-cortex stimulation can trace recognizable letter-like forms in sighted and blind participants [23]. Cellular-resolution optogenetic stimulation of tuned ensembles can recruit related populations and bias orientation discrimination, with different layer-specific thresholds [24]. Triggering pattern-completion neurons can recall behaviorally relevant ensembles and alter choices [25]. Perceptography can infer complex stimulation-linked changes in macaque perception and shows that those changes depend on concurrent input, site, and intensity [26].

Yet targeting visually selective cells is not always the best predictor of behavioral influence. A 2025 study found that the stimulation-induced change in the surrounding network predicted perceptual impact better than the selectivity of the stimulated ensemble [27]. Local recurrent-circuit perturbation also shows why excitation and suppression must be resolved by distance and tuning rather than compressed into generic recurrence [28]. These results make the central test sharper: the evidentiary variable is not merely *what a cell decodes*. It is what the intervention causes the larger receiver network to do.

3 The SAN genealogy of the problem

The present model is a 2026 synthesis, but its component questions have a dated history. An internally dated May 2011 note described a stereographic-camera analogy in which distributed “pixels” contribute different

portions of a changing scene, learned statistical relations support inference and prediction, and the senses contribute to a continuously updated multisensory picture. Its 2011 date is an owner record; immutable public Git custody is later.

The public 2017 Neural Lace material made the problem concrete. In *The Neural Lace Podcast* episode 4, hosted by Micah Blumberg with guest Andre Watson and published on May 5, the discussion first states the invariance problem: one does not encounter exactly the same neural signal or photograph twice, yet one still recognizes an object. It then moves from local lines, edges, and colors through leaves and branches to a tree, and from a bottle, glass, or table to distinguishable spatial-temporal neural-wave patterns. Finally, it asks what happens if a computer writes a glass-associated pattern into a brain that currently sees no glass: where would the glass appear, how would it interact with the current scene, and could a badly placed write produce hallucination or destabilize reality? These propositions are bound to the local transcript at lines 419-591 and to the public video route <https://youtu.be/RRN91RwiYiw>; because the current transcript has line but not audio-time locators, The source-locator, data-contract, and benchmark specification paraphrases rather than manufacturing timestamps or speaker precision beyond the transcript’s host/guest identification.

The public June 2017 episode 7 and its accompanying article add the closed-loop answer. The episode introduction states that a future interface should first listen to the pattern present in the current person and environment and then send only the difference between the present and desired pattern. It separates three interface modes - emotion/intention inference, task or game control, and direct EEG-driven light, sound, or tactile feedback - before the later discussion describes EEG-driven 3D geometry as a learned mirror through which a participant forms concepts about changes in their own brain activity. These propositions are bound to nlp7.md lines 25-90 and 470-516 and to the June 7 public SoundCloud route. The independently accessible article *The Brain as a Special Kind of Hard Drive* was published on Medium on June 11, with an internal June 7 date, and is now also rendered at svgn.io; the public page repeats the episode route, the three interface modes, and the read/write proposal. The tongue-strip example emphasizes that a camera signal delivered through touch can acquire spatial use through learning, consistent with the bounded sensory-substitution literature [29-32]. Together these sources establish the 2017 problem and engineering ancestry, while leaving the named PWD and complete qualitative-rendering operator to their later dated stages.

The 2022 public Git corpus named a more explicit architecture. The initial a0001z formulation joined distributed phase changes, reference frames, point-cloud semantic segmentation, and self-aware rendering. a0039z described arrays that both contribute and listen, with multisensory spatial-temporal correlations and topographic phase differences. a0238z explicitly compared object segmentation and classification with 3D semantic segmentation and rejected a single-neuron viewer. a0149z described learned contours and properties in a 3D phase topology received by downstream structures. a0269z used the functional analogy of one circuit’s pattern serving as a display for another. The August and September a0001z stages refined the fine-grained grid analogy and described redness as a receiver-detectable difference from tonic activity that could persist through recurrence.

The 2024 SA0v9 heading “3D Semantic Segmentation to 3D Qualia” accurately names the intended bridge but contains only a program outline. The current paper earns the newer term *qualitative neural rendering* only after specifying the intervening operations and their tests.

4 The operation before the name

The proposed operation can be stated without SAN vocabulary.

First, different sensory populations supply partial measurements. Second, local and distributed processes

estimate classes, instances, surfaces, borders, depth, and motion while retaining uncertainty. Third, candidate parts are transformed among modality-, body-, and world-relative frames so that signals can be tested for common object membership. Fourth, each receiving population transforms the arriving pattern according to its existing state and learned sensitivities. Fifth, recurrent exchange stabilizes, splits, merges, suppresses, or revises the object hypothesis. Sixth, the updated distributed state changes recognition, report, choice, and action. Seventh, movement and new sensory evidence return to test the model.

The resulting state is not a finished picture sent to a separate observer. The receiver populations are altered by the organized differences, and those alterations change what the network can do next. The “viewer” operation is the network’s consequential reception, transformation, and recurrence.

This complete candidate operation is called **qualitative neural rendering**: the distributed process by which object-wise, boundary-wise, multisensory constraints become receiver-relative neural differences with content-specific recurrent and behavioral consequences. The term *qualitative* marks the target - differentiated content - not an assumption that the target has already been proved.

Within SAN, a **Phase Wave Differential** names the receiver-relative departure between an arriving, expected, or ongoing phase-pattern relation and the state actually registered by the receiver. A PWD is more specific than a traveling wave or generic phase code because it requires a declared sender, receiver, delay, tonic expectation, observed difference, and consequence. It enters this paper after the problem and operation are already clear.

5 Formal specification

5.1 Partial observations and object variables

Let the observations available at time $t \in \mathbb{R}_{\geq 0}$ be a typed collection across modalities $m \in \mathcal{M}$:

$$\mathcal{X}_t = \{x_i^{(m)}(t), \ell_i^{(m)}, \tau_i^{(m)}, u_i^{(m)}\}_{m,i}, \quad (1)$$

where $x_i^{(m)} \in \mathcal{X}^{(m)}$ is the measured feature, $\ell_i^{(m)} \in \mathcal{L}^{(m)}$ its sensor or neural location, $\tau_i^{(m)}$ its acquisition time in seconds relative to a declared clock, and $u_i^{(m)} \geq 0$ its uncertainty in the units of the associated estimator. A semantic estimator returns a distribution over classes $k \in \mathcal{K}$, not only a hard label:

$$q_i^{(m)}(k, t) = P(K_i = k \mid x_i^{(m)}(t), \theta_q). \quad (2)$$

Instance membership is a separate latent assignment:

$$z_i^{(m)}(t) \in \{1, \dots, N_t, \emptyset\}, \quad (3)$$

so two points may share a semantic class while belonging to different objects. For adjacent elements (i,j), border ownership is a signed or probabilistic relation:

$$b_{ij}^{(m)}(t) = P(i \leftarrow \partial_{ij} \mid \mathcal{X}_t) - P(j \leftarrow \partial_{ij} \mid \mathcal{X}_t), \quad (4)$$

where positive and negative values indicate opposite ownership directions. The object hypothesis o at time t is therefore not a label but a typed tuple:

$$H_o(t) = (K_o, Z_o, \partial_o, G_o, V_o, U_o)_t, \quad (5)$$

containing class, instance membership, boundary, geometry, visibility or occlusion state, and uncertainty.

5.2 Multisensory registration

Each modality is mapped through a receiver- and time-dependent transform into a declared reference frame f :

$$y_i^{(m \rightarrow f)}(t) = T_{m \rightarrow f} \left(x_i^{(m)}(t); p_t, r_f(t) \right), \quad (6)$$

where p_t includes eye, head, limb, and body configuration and r_f is the receiving population's state. No equation requires one universal frame. Pairwise registration error for hypothesized common object o is

$$E_{\text{reg}}(o, t) = \sum_{m < n} \sum_{i \in o} \rho \left(y_i^{(m \rightarrow f)}(t), y_{\pi(i)}^{(n \rightarrow f)}(t) \right) w_{imn}, \quad (7)$$

with robust distance ρ_f defined in the destination frame f , correspondence π_{mn} , and dimensionless reliability weight $w_{imn} \in [0, 1]$. A metric distance is not evaluated until the source and destination frames, scale, and units are declared. Object membership should improve a joint energy that trades data fit, boundaries, cross-modal consistency, and temporal continuity:

$$\mathcal{E}(H_t) = \lambda_d E_{\text{data}} + \lambda_b E_{\text{border}} + \lambda_r E_{\text{reg}} + \lambda_c E_{\text{continuity}}. \quad (8)$$

5.3 Tonic receiver state and PWD

For receiver j , define a tonic conditional expectation and covariance from comparable preceding states:

$$\mu_j(t) = \mathbb{E}[s_j(t) \mid c_t], \quad \Sigma_j(t) = \text{Cov}[s_j(t) \mid c_t], \quad (9)$$

where c_t includes task, body, recent history, oscillatory phase, and other declared context. The arriving sender pattern $a_{ij}(t - d_{ij})$ is transformed through the receiver:

$$\hat{s}_{ij}(t) = F_j(a_{ij}(t - d_{ij}), r_j(t), W_{ij}, I_j(t)), \quad (10)$$

where W_{ij} is learned coupling and I_j is inhibition and modulatory state. The receiver-relative difference is

$$\delta_{ij}(t) = \hat{s}_{ij}(t) - \mu_j(t). \quad (11)$$

A scale-aware magnitude can be measured by

$$D_{ij}^2(t) = \delta_{ij}(t)^\top (\Sigma_j(t) + \varepsilon_\Sigma I)^{-1} \delta_{ij}(t), \quad (12)$$

while conditional self-information follows the declared conditional probability model [33,35]:

$$\mathcal{I}_{ij}(t) = -\log_2 p(\delta_{ij}(t) \mid c_t, H_o(t)). \quad (13)$$

Neither distance nor surprisal alone supplies biological meaning. A PWD becomes operationally relevant only when it changes the receiver and downstream trajectory:

$$r_j(t+1) = G_j(r_j(t), \delta_{ij}(t), H_o(t), I_j(t)). \quad (14)$$

5.4 Recurrent object revision and content

Recurrent exchange revises each object hypothesis using incoming evidence, other hypotheses, receiver state, and the previous hypothesis:

$$H_o(t+1) = \mathcal{R}_o(H_o(t), \mathcal{X}_{t+1}, \{r_j(t+1)\}, \{H_{o'}(t)\}_{o' \neq o}). \quad (15)$$

A split, merge, or persistence decision can be expressed as the model choice

$$M_o^* = \arg \min_{M \in \{\text{persist, split, merge, new, absent}\}} [\mathcal{E}(H_t \mid M) + \beta \Omega(M)], \quad (16)$$

where Ω penalizes unnecessary complexity without prohibiting revision. The distributed content state is a typed set rather than one scalar:

$$C_t = \left\{ H_o(t), r_j(t), \delta_{ij}(t), A_t, B_t \right\}_{o,i,j}, \quad (17)$$

where A_t denotes accessible action or report policies and B_t denotes relevant body-state variables. Two trials differ in predicted content when their object organization and downstream affordances differ after matched low-level input controls:

$$d_C(C_t, C'_t) = w_H d_H + w_r d_r + w_\delta d_\delta + w_A d_A + w_B d_B. \quad (18)$$

This is a measurement proposal, not a claim that a weighted distance exhausts experience.

5.5 Action-return closure

The current state changes the distribution over possible actions:

$$\pi(a_t \mid C_t) = \text{softmax}(V(C_t, a_t)/\tau). \quad (19)$$

Action changes the world and produces new evidence. The returned mismatch is

$$e_{t+1} = \mathcal{X}_{t+1} - \hat{\mathcal{X}}_{t+1}(H_t, a_t, p_t), \quad (20)$$

which can preserve, revise, split, merge, or reject the prior object hypothesis through Equation [15]. Qualitative neural rendering is therefore a closed estimation-and-action process, not a terminal image.

5.6 Incremental and causal evidence

Let $M_0 \dots M_5$ denote increasingly complete models: low-level/feedforward; semantic; instance-plus-border; recurrent completion; multisensory registration; and receiver-state/PWD. Incremental held-out value of the full model is

$$\Delta_{5:4} = \mathcal{L}_{\text{test}}(M_4) - \mathcal{L}_{\text{test}}(M_5), \quad (21)$$

where positive values indicate reduced loss and all model-selection decisions are frozen before final evaluation. A causal effect of intervention u on content outcome Y_o is stated using intervention notation [36]:

$$\text{ACE}_{u,o} = \mathbb{E}[Y_o \mid \text{do}(u = 1)] - \mathbb{E}[Y_o \mid \text{do}(u = 0)]. \quad (22)$$

Content specificity requires the predicted target effect to exceed matched off-target and nonspecific effects:

$$S_{u,o} = |\text{ACE}_{u,o}| - \max(|\text{ACE}_{u,o'}|, |\text{ACE}_{u,\text{salience}}|, |\text{ACE}_{u,\text{motor}}|). \quad (23)$$

Matched rescue is quantified by

$$R_{u,o} = 1 - \frac{|Y_o^{\text{rescue}} - Y_o^{\text{baseline}}|}{|Y_o^{\text{perturb}} - Y_o^{\text{baseline}}| + \epsilon}. \quad (24)$$

A strong result requires positive incremental value, target-specific perturbation, and selective rescue with uncertainty intervals - not one impressive decoding score.

5.7 Domains, units, estimators, and limiting cases

The equations above become executable only after their domains and units are frozen. Class probabilities and reliability weights are dimensionless. Spatial positions may be measured in pixels, millimeters, degrees of visual angle, or body-normalized coordinates, but two positions cannot enter one distance until an explicit transform and conversion place them in the same frame and unit. Phase is measured in radians on the circle; time and latency are measured in seconds; firing rate is measured in spikes per second; field potential and stimulation intensity retain their acquisition-specific physical units. Learned embeddings are unitless coordinates and must never be silently treated as anatomical distance.

Log loss and surprisal require probabilities bounded away from zero. For a preregistered calibration floor $0 < \varepsilon_p < 1/2$, evaluation uses

$$\tilde{p}_k = \frac{\max(\varepsilon_p, \min(1 - \varepsilon_p, p_k))}{\sum_{h \in \mathcal{K}} \max(\varepsilon_p, \min(1 - \varepsilon_p, p_h))}, \quad (25)$$

and reports a sensitivity analysis over the frozen admissible values of ε_p . Singular or poorly conditioned receiver covariance is handled by a declared shrinkage estimator,

$$\widehat{\Sigma}_{j,\lambda} = (1 - \lambda)\widehat{\Sigma}_j + \lambda \frac{\text{tr}(\widehat{\Sigma}_j)}{d_j} I, \quad 0 < \lambda \leq 1, \quad (26)$$

where d_j is receiver-state dimension and λ is selected only on development data. Equation [12] is undefined when the resulting matrix is not positive definite; such a trial is marked invalid rather than repaired after seeing the target result.

For registration, The source-locator, data-contract, and benchmark specification freezes the Huber loss in the declared destination frame,

$$\rho_f(a, b) = \begin{cases} \frac{1}{2} \|a - b\|_f^2, & \|a - b\|_f \leq \kappa_f, \\ \kappa_f (\|a - b\|_f - \frac{1}{2} \kappa_f), & \|a - b\|_f > \kappa_f, \end{cases} \quad (27)$$

with κ_f expressed in the same unit as the coordinates. Transform composition is permitted only when adjacent codomain and domain types agree:

$$T_{a \rightarrow c} = T_{b \rightarrow c} \circ T_{a \rightarrow b}, \quad \text{codom}(T_{a \rightarrow b}) = \text{dom}(T_{b \rightarrow c}). \quad (28)$$

The proposed content distance in Equation [18] is a weighted dissimilarity, not automatically a metric. Each component is normalized by a development-set scale $s_v > 0$,

$$\tilde{d}_C(C, C') = \sum_{v \in \{H, r, \delta, A, B\}} w_v \frac{d_v(v, v')}{s_v}, \quad w_v \geq 0, \quad \sum_v w_v = 1, \quad (29)$$

and may be called a metric only if every active d_v satisfies non-negativity, identity, symmetry, and the triangle inequality. Otherwise the manuscript uses *dissimilarity*.

The rescue score in Equation [24] is not interpretable when the perturbation has no resolvable target effect. Let $\tau_Y > 0$ be a preregistered minimum effect and $CI_{1-\alpha}(\Delta_u)$ the clustered uncertainty interval. Rescue is admissible only when

$$A_{u,o} = \mathbf{1}(|\widehat{\Delta}_{u,o}| \geq \tau_Y \wedge 0 \notin CI_{1-\alpha}(\Delta_{u,o})) = 1. \quad (30)$$

If $A_{u,o} = 0$, $R_{u,o}$ is NOT-IDENTIFIED, not zero, one, or evidence of recovery. If admissible, the score and its bootstrap or hierarchical interval are reported together; values below zero indicate worsening and values above one indicate overshoot rather than being clipped into a favorable range. Missing observations, excluded trials, and failed manipulations are counted separately from outcome values.

5.8 Identifiability and counterexamples

Prediction alone cannot identify the proposed mechanism. Two models M_a and M_b are observationally equivalent on environment family $\mathcal{E}$ when

$$P_{M_a}(Y \mid X, e) = P_{M_b}(Y \mid X, e) \quad \text{for every } e \in \mathcal{E}, \quad (31)$$

even if their hidden states carry different names. Identification requires a frozen context or intervention q for which their distributions separate by more than a declared tolerance $\eta > 0$:

$$\sup_{q \in \mathcal{Q}} D(P_{M_a}(Y \mid do(q)), P_{M_b}(Y \mid do(q))) > \eta, \quad (32)$$

where D and η are chosen before the final test. Equal performance leaves the mechanism unresolved.

Concrete counterexamples keep the model ladder honest. M1 can give the same broccoli class to two florets and therefore cannot preserve their identities. M2 can preserve those identities but cannot decide whether two disconnected regions are one object behind an occluder. M3 can complete the object while placing touch and sound in the wrong body/world frame. M4 can register all modalities yet predict the same outcome when prestimulus receiver state changes. M5 can fit that state dependence without identifying a PWD if a global-gain or generic latent-state model makes the same interventional predictions. These are not rhetorical weaknesses; they define the exact contrast each subsequent model must earn.

5.9 Normalized objectives and model-choice discipline

Equation [8] is not executable if one term grows merely because it contains more observations or uses a larger numerical unit. For each component $q \in \{d, b, r, c\}$, let $e_{q\ell}$ be its per-observation loss, $n_q > 0$ its valid observation count, and $s_q > 0$ a robust scale estimated on development data only. Define

$$\overline{E}_q = \frac{1}{n_q} \sum_{\ell=1}^{n_q} \frac{e_{q\ell}}{s_q}, \quad (33)$$

and replace an untyped sum by the dimensionless joint objective

$$\mathcal{E}_{\text{norm}}(H_t) = \sum_{q \in \{d, b, r, c\}} \lambda_q \overline{E}_q, \quad \lambda_q \geq 0, \quad \sum_q \lambda_q = 1. \quad (34)$$

The scales and weights are frozen before the final test. A zero-observation component is not silently assigned zero loss; it is marked unavailable, and the trial follows the preregistered missing-component rule. Sensitivity analyses vary the development-set scales and weights without using final outcomes to select a favorable combination.

Equation [16] also requires a fixed complexity rule. For model M with development loss $\mathcal{L}_{\text{dev}}(M)$, p_M optimized degrees of freedom, and effective development sample size $n_{\text{eff}} > 1$, The mathematics, estimator, and identifiability audit uses the declared per-observation score

$$\text{Score}(M) = \mathcal{L}_{\text{dev}}(M) + \frac{p_M \log n_{\text{eff}}}{2n_{\text{eff}}}. \quad (35)$$

This is a benchmark contract, not a claim that one penalty is universally correct. The same frozen score is applied to every member of the ladder; parameter sharing, early stopping, hyperparameter search, and

representation pretraining are counted in a separately reported resource ledger. A second analysis matches parameter count and optimization exposure directly. A richer model earns promotion only if its final held-out result survives both views.

5.10 Receiver-update identification and action calibration

The receiver update in Equation [14] can be confused with ordinary state persistence. The mathematics, estimator, and identifiability audit therefore estimates a baseline update from pretreatment variables that exclude the candidate PWD. With cross-fitting, let

$$u_j(t) = r_j(t+1) - \widehat{G}_{0,-k}(r_j(t), H_o(t), I_j(t), c_t) = B_j \delta_{ij}(t) + \xi_j(t), \quad (36)$$

where $\widehat{G}_{0,-k}$ is trained without fold k , and u_j is evaluated on that held-out fold. The coefficient or nonlinear analogue associated with δ_{ij} is not interpreted causally from observation alone. Identification requires either randomized variation, a justified instrument, or the intervention contract below. Context variables are chosen before treatment; post-intervention activity cannot be used to erase the effect being estimated.

The action temperature in Equation [19] is estimated on development trials within a frozen positive interval:

$$\tau^* = \arg \min_{\tau \in [\tau_{\min}, \tau_{\max}]} - \sum_{(C,a) \in \mathcal{D}_{\text{dev}}} \log \pi_{\tau}(a \mid C). \quad (37)$$

Final-test choices, confidence reports, and action outcomes are evaluated with τ^* fixed. The benchmark reports whether conclusions persist across a declared neighborhood of temperatures, because an arbitrarily low temperature can manufacture apparent certainty and an arbitrarily high temperature can erase meaningful action differences.

5.11 Divergence, tolerance, and causal assumptions

The mathematics, estimator, and identifiability audit instantiates the divergence in Equation [32] as base-two Jensen-Shannon divergence unless a preregistered outcome requires another proper divergence. For predictive distributions P and Q , with $M = (P + Q)/2$,

$$D_{\text{JS}}(P, Q) = \frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M), \quad 0 \leq D_{\text{JS}} \leq 1 \text{ bit}. \quad (38)$$

The separation tolerance is not chosen after seeing a target comparison. Let $D_{\text{null}}^{(b)}$ denote divergences generated by label-preserving permutations, duplicated-model seeds, or another frozen null procedure. Then

$$\eta = \max \left(\eta_{\text{practical}}, Q_{1-\alpha} \{ D_{\text{null}}^{(b)} \}_{b=1}^B \right), \quad (39)$$

where $\eta_{\text{practical}} > 0$, α , B , and the null generator are fixed before final testing. A gap below η is unresolved, not evidence that two mechanisms are identical.

For randomized perturbation $U \in \{0, 1\}$, potential outcomes must obey a declared intervention and interference map. Within each randomization block, the identifying assumptions are expressed as

$$(Y_o(1), Y_o(0)) \perp U \mid C_{\text{pre}}, \quad 0 < P(U = 1 \mid C_{\text{pre}}) < 1, \quad (40)$$

along with consistency between the assigned and received intervention. Neural systems contain interference by design, so the experimental unit cannot be presumed independent at the single-neuron level. The protocol instead declares the ensemble, region, participant, session, or scene cluster within which interference is allowed and randomizes across valid clusters. Failed manipulation checks, spillover beyond the declared map, and post-treatment conditioning are reported rather than repaired away.

5.12 Distinguishing a receiver-relative PWD from global gain and a generic latent state

A global-gain account predicts that a shared scalar g_t changes many channels together. A receiver-relative account predicts an additional relation between the particular receiver and the arriving difference. One diagnostic model is

$$Y_o = \beta_0 + \beta_g g_t + \beta_\delta D_{ij} + \beta_j \mathbf{1}[j = j^*] + \beta_{\delta j} D_{ij} \mathbf{1}[j = j^*] + \gamma^\top C_{\text{pre}} + \epsilon, \quad (41)$$

where j^* is the prospectively named receiver and D_{ij} is Equation [12]. The interaction is a discriminator, not by itself a causal proof. The decisive comparison combines receiver-matched and receiver-mismatched interventions, holds total stimulation or activation magnitude fixed, and tests whether the predicted target content and downstream route change selectively.

The generic latent comparator receives the same observation history, prestimulus state, parameter budget, optimization budget, and random seeds as M5, but it is not given the typed sender, receiver, delay, tonic expectation, or PWD relation. Let M_G be the matched global-gain model and M_L the generic latent-state model. The minimum adjusted advantage of M5 is

$$\Gamma_5 = \min\{\text{Score}(M_G) - \text{Score}(M_5), \text{Score}(M_L) - \text{Score}(M_5)\}. \quad (42)$$

M5 is not called identified unless Γ_5 exceeds its frozen practical threshold on held-out scenes and the interventional distributions separate under Equations [38]-[40]. If a generic latent model matches every intervention, the receiver-relative interpretation remains unseparated even when M5 predicts well.

5.13 Dependence-aware budget, numerical precision, timescale, and stability

Repeated trials from one participant, scene, network seed, or recording session are not independent observations. For N total observations arranged in equal clusters of size m with planning intraclass correlation ρ_c , an approximate planning quantity is

$$N_{\text{eff}} = \frac{N}{1 + (m - 1)\rho_c}. \quad (43)$$

The final analysis uses the actual hierarchical design rather than treating Equation [43] as a substitute for it. Power is estimated by simulation across a frozen grid of plausible effect sizes, base rates, cluster correlations, missingness, and manipulation failures. The manuscript reports the entire assurance curve and does not choose one optimistic point estimate.

For a simulated rejection or success proportion $\hat{p}$ across B independent seeds, its Monte Carlo standard error is

$$\text{MCSE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{B}}. \quad (44)$$

Simulation continues until the worst-case MCSE over the preregistered planning grid is below its tolerance or the hard budget is reached; hitting the budget without precision produces INDETERMINATE-BUDGET, not a favorable extrapolation.

The receiver and recurrent update must also be dynamically interpretable. For a fixed-point analysis over a declared update interval, local stability requires

$$\rho(J_{\mathcal{G}}(x^*)) < 1, \quad J_{\mathcal{G}} = \frac{\partial \mathcal{G}}{\partial x}, \quad (45)$$

where ρ is spectral radius and $\mathcal{G}$ is the combined receiver-object update. This condition is not imposed on every oscillatory trajectory. A stable limit cycle is assessed through its return map, and transient amplification is reported through finite-time gain. Acquisition, sender-receiver delay, recurrent revision, action selection, and returned sensory evidence retain separate timestamp fields; The mathematics, estimator, and identifiability audit does not collapse them into one undifferentiated time index.

5.14 Worked sanity checks and countermodels

The following calculations are deterministic checks of definitions, not benchmark or biological results.

Target	Frozen toy input	Required result or disposition
Equation [25]	$p = (0, 0.7, 0.3), \varepsilon_p = 0.01$	$\tilde{p} \approx (0.009901, 0.693069, 0.297030)$; entries remain positive and sum to one.
Equation [26]	$\hat{\Sigma} = \text{diag}(4, 0), d = 2, \lambda = 0.1$	$\hat{\Sigma}_\lambda = \text{diag}(3.8, 0.2)$, which is positive definite.
Equation [27]	residual norm 2, $\kappa_f = 1$	Huber loss 1.5, rather than squared loss 2.
Equation [28]	retinal-to-head then head-to-world transforms	Composition is valid; retinal-to-head followed by hand-to-world is rejected as a type mismatch.
Equation [29]	normalized components 0.5, 1.0, weights 0.4, 0.6	dissimilarity 0.8; no metric claim is inferred from the weighted sum alone.
Equation [30]	effect 0.20, threshold 0.10, interval $[0.05, 0.35]$	rescue admissible; effect 0.04 with interval $[-0.02, 0.10]$ returns NOT-IDENTIFIED.
Equations [31]-[32], [38]	observed Bernoulli probability 0.7 for both models; intervention probabilities 0.2 versus 0.8	observationally equivalent on the observed family, but intervention $D_{JS} \approx 0.2781$ bits.
Equations [33]-[34]	six data losses sum to 12 at scale 2; two border losses sum to 4 at scale 2	both normalized components equal 1; count alone cannot dominate the objective.
Equation [35]	$n_{\text{eff}} = 1000$; models $(\mathcal{L}, p) = (0.40, 20)$ and $(0.37, 50)$	scores approximately 0.4691 and 0.5427; the lower raw loss does not automatically win.
Equation [36]	observed receiver update 0.7, cross-fit baseline 0.5, candidate $\delta = 0.4$	residual update 0.2, compatible with $B = 0.5$, but not causal without the intervention assumptions.
Equation [39]	practical threshold 0.03, 95th null quantile 0.04	$\eta = 0.04$.
Equation [42]	scores $M_5 = 0.31, M_G = 0.34, M_L = 0.33$	$\Gamma_5 = 0.02$; promotion still depends on the frozen practical threshold and interventions.
Equation [43]	$N = 240, m = 12, \rho_c = 0.1$	$N_{\text{eff}} \approx 114.29$, demonstrating why 240 repeated trials are not 240 independent units.
Equation [44]	$\hat{p} = 0.5, B = 10,000$	MCSE 0.005.
Equation [45]	spectral radius 0.82 versus 1.07	first fixed point passes the local test; second fails it without implying a global statement about all dynamics.

Eight countermodels are carried forward as mandatory interventions: same class but different instances; fixed pixels with reversed border ownership; one occluded object versus two disconnected objects; matched content with a cross-modal displacement; identical external input under different receiver states; receiver-matched versus receiver-mismatched perturbation at fixed total magnitude; capacity-matched global gain; and a generic latent-state model plus matched attention, report, eye movement, and action preparation. Any implementation that omits one of these contrasts cannot claim to have run the full The mathematics, estimator, and identifiability audit identification contract.

5.15 Calibration, distribution shift, ablation, and replication gates

A bounded score is not automatically a calibrated probability. Partition development-frozen predictions into B bins I_b . For bin membership set S_b , observed positive frequency $\text{freq}(S_b)$, and mean reported probability $\text{conf}(S_b)$, expected calibration error is

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{N} |\text{freq}(S_b) - \text{conf}(S_b)|. \quad (46)$$

ECE is reported with accuracy and Brier score rather than used to relabel the sigmoid-wrapped ridge margin as a biological probability. Calibration can change under distribution shift even when ranking performance changes little [39,40].

For seed k , shift scenario s , typed model M_5 , and comparator M_c , the paired receiver-task accuracy gap is

$$\Delta_{csk} = A_{5sk} - A_{csk}. \quad (47)$$

Every nonpositive value is retained. The scenario mean is not sufficient by itself; the paper reports the seed distribution, independent replication, and the worst scenario for each model:

$$W_m = \min_{s \in \mathcal{S}} \frac{1}{K_s} \sum_{k=1}^{K_s} A_{msk}. \quad (48)$$

Typed-feature necessity is tested without reducing fitted dimension. Let $M_{5 \setminus q}^{\text{sham}}$ replace typed channel q by a seed-fixed noise channel in training and validation while preserving all other inputs, optimization, and parameter count. Its paired gap is

$$\Lambda_{qsk} = A_{5sk}^{(q)} - A_{5 \setminus q, sk}^{\text{sham}, (q)}. \quad (49)$$

A positive Λ shows dependence on that input in this fitted synthetic system; it does not prove the biological interpretation assigned to the input. Conversely, a nonpositive sham gap is retained as evidence that the target is recoverable from correlations or confounds under that scenario.

The distribution-shift and latent-baseline stress test therefore defines a conjunctive development gate rather than selecting the most favorable mean:

$$G_6 = \mathbf{1} \left[\min_{c,s} \bar{\Delta}_{cs} > \delta_{\min} \right] \mathbf{1} \left[\max_s \text{ECE}_{5s} < \epsilon_{\max} \right] \mathbf{1} \left[\min_{q,s} \bar{\Lambda}_{qs} > \lambda_{\min} \right] \mathbf{1} [\text{Replicate}(\Delta, \Lambda) = 1]. \quad (50)$$

The thresholds and multiplicity rule must be frozen before final testing. The distribution-shift and latent-baseline stress test does not pass this gate: latent comparators equal or exceed M5 in some shifted scenarios, at least one sham gap nearly collapses under confound reversal, and calibration error exceeds any plausible stringent threshold. The failed conjunction is the scientifically useful outcome because it identifies which controls and estimators require improvement before a biological claim can be tested.

6 Six discriminating hypotheses

H1 - Object-organization increment. Instance identity and border ownership will predict held-out object-content judgments beyond low-level features and semantic class alone, especially under occlusion and shared-border ambiguity.

H2 - Recurrence scales with segmentation demand. Disrupting recurrent processing will have a larger and later content-specific effect when background complexity, occlusion, or border ambiguity increases, while leaving an earlier local-feature signal comparatively intact.

H3 - Registration is content bearing. When visual, tactile, auditory, and proprioceptive signals plausibly refer to one object, cross-modal registration variables will predict object-location and ownership reports beyond modality-specific decoders. Controlled misregistration will produce displacement or misbinding rather than only reduced confidence.

H4 - Receiver state matters. The same physical input and nominal segmentation will lead to different recognized content as a function of ongoing receiver state. The M5 receiver-state/PWD model will improve held-out prediction beyond M4 without merely absorbing report criterion or global arousal; sensitivity and response criterion are analyzed separately [34].

H5 - Network consequence outranks selectivity alone. The content-specific effect of targeted stimulation will be predicted more accurately by induced receiver-network trajectory than by the target cells' stimulus selectivity alone, consistent with the need to model downstream reception [27].

H6 - Causal object transformation. A valid intervention at a predicted object/border/registration route will create a specific split, merge, displacement, boundary reversal, or feature misbinding, and the matched inverse or compensating intervention will rescue that effect. Generic salience, phosphene, confusion, response bias, or motor disruption will not count as the target result.

7 Object-Boundary Registration Benchmark

7.1 Application purpose

The first application is a reproducible benchmark rather than an artificial claim of machine consciousness. It tests whether the typed variables and equations add explanatory and predictive value across controlled scenes. Synthetic 3D environments permit independent manipulation that natural images often confound; natural scenes test whether success survives realism.

Each scene family independently varies semantic class, instance count, boundary ownership, depth, occlusion, viewpoint, lighting, object motion, observer motion, action, cross-modal correspondence, and uncertainty. A broccoli family, for example, can hold texture constant while changing whether two florets are one partly occluded instance or two instances; can move a border without changing class; can pair congruent or displaced touch and sound; and can reveal hidden geometry through action.

7.2 Model ladder

- **M0:** low-level and feedforward features only.
- **M1:** M0 plus semantic class distributions.
- **M2:** M1 plus instance identity and border ownership.
- **M3:** M2 plus recurrent completion and temporal continuity.
- **M4:** M3 plus multisensory and body/world reference-frame transforms.
- **M5:** M4 plus tonic receiver state, receiver-relative PWD, and downstream network consequence.
- **MG:** capacity-matched M4 plus a global scalar gain applied without a typed sender-receiver relation.
- **ML:** capacity-matched M4 plus a generic latent state supplied the same observation history but no explicit PWD relation.

Every higher model must inherit the lower model's inputs so that improvement cannot be attributed to privileged raw data. Parameter counts, optimization budgets, training exposure, preprocessing, early stopping, and search exposure must be matched or entered into the resource ledger. M5 must outperform not

only M4 but also MG and ML under the adjusted score and intervention contract; otherwise receiver-relative identification remains open.

7.3 Tasks and metrics

Computational tasks include semantic intersection-over-union, instance average precision, boundary F1, border-ownership area under the receiver-operating curve, 3D registration error, occluded-shape completion, held-out object identity, object-location report, action success, and calibration. Neural tasks add time-resolved decoding, cross-condition generalization, representational similarity, state-conditioned prediction, and mediation by induced network trajectory.

No aggregate score can conceal a failed load-bearing dimension. Results are reported as a vector containing class, instance, boundary, registration, recurrence, receiver-state increment, causal specificity, rescue, uncertainty, and cross-scene generalization.

The final-test budget is not selected from a single guessed variance. A development-only simulation crosses effect size, class balance, instance count, occlusion, cluster correlation, missingness, perturbation uptake, and model-seed variability. For each primary contrast, the release reports planned clusters, trials per cluster, raw trials, Equation [43] effective size, assurance, multiplicity rule, MCSE, expected failures, and hard stopping limit. A favorable point estimate without its dependence-aware budget is not a completed result.

7.4 Controlled experiments

Experiment 1: synthetic factorization. Train and evaluate M0-M5 on seeded scenes with fully known class, instance, boundary, depth, and transform variables. Hold out combinations of viewpoint, occlusion, motion, and action. Test whether each added variable performs its intended job rather than merely increasing capacity.

Experiment 2: recurrent scene segmentation. Present humans with isolated and embedded objects across graded background complexity and occlusion. Use masking or carefully timed perturbation to distinguish early feedforward feature availability from later segmentation-dependent processing. Record behavior with confidence and signal-detection measures.

Experiment 3: ambiguous ownership. Use bistable or matched-edge displays whose local statistics remain constant while perceived ownership changes. Separate stimulus, report, attention, eye movement, and motor preparation through factorial task design and no-report-compatible measures.

Experiment 4: multisensory registration. Pair visual, tactile, auditory, and proprioceptive evidence under congruent and parametrically displaced body-centered and world-centered conditions. Predict when the percept should bind, split, or move and test whether reference-frame variables improve held-out predictions.

Experiment 5: perturbation and rescue. In an ethically appropriate preparation, perturb a boundary, object-completion, or registration route at the predicted time. Predefine target outcomes such as boundary reversal, feature migration, object split, object merge, or spatial displacement. Apply a matched rescue that restores the predicted relation without simply increasing global activation.

Experiment 6: staged neural writing. Begin with established simple-form stimulation and advance only after safety and specificity gates. Compare static, sequential, phase-structured, and receiver-state-conditioned patterns. Measure what the surrounding network does and whether the effect integrates with current sensory evidence. A complete steak-and-broccoli percept is a distant engineering target, not the

entry criterion.

7.5 Reproducibility contract

The benchmark freezes scene generators, seeds, splits, preprocessing, model capacity rules, hypotheses, primary outcomes, exclusion criteria, and analysis code before final tests. It preserves raw event logs, intervention schedules, fitted parameters, and negative results. Each result must be traceable from figure cell to raw trial and from raw trial to protocol version and content hash. Error thresholds, multiplicity handling, uncertainty, and replication obligations are declared prospectively rather than chosen from the observed outcome [37,38].

The source-locator, data-contract, and benchmark specification implemented the data-contract layer before building a model. The mathematics, estimator, and identifiability audit added the estimator, tolerance, causal-assumption, comparator, sample-budget, numerical-precision, and stability contracts that the application must consume. The versioned event schema keeps class, instance, surface, directed border, source and destination frames, receiver state, intervention, outcome, and provenance separate. Six golden edge cases contribute thirteen traceable events: same-class/different-instance, reversed ownership with the local edge fixed, increased occlusion with identity fixed, cross-modal displacement, receiver-state change with physical input fixed, and a null perturbation for which rescue is explicitly NOT-IDENTIFIED. The deterministic validator and six adversarial regression tests pass. The executable reference application implements the typed reference application and leaves the final-test seal unopened. These are software-contract, synthetic fixture, and mathematical sanity-check results, not trained M0-M5 performance, biological data, or evidence of consciousness.

7.6 Executable reference-application result

The executable reference application supplies an executable path from the equations to traceable events. The implementation is deliberately small and auditable: a typed `Scene` defines class, instance, border owner, object count, cross-modal offset, receiver condition, target and actual receiver, typed route, generic state, global activation, four confounds, visible regions, and time. A typed `Prediction` keeps the corresponding model outputs, budget, and local stability radius separate. The generator produces paired scenes for the eight countermodels. Each pair has a prospectively declared change; the factor monitor compares every dataclass field and fails if an undeclared scientific variable changes.

The reference ladder makes each burden executable without pretending that a hand-authored implementation is an empirical comparison. M0 and M1 lack explicit instance and border variables; M2 adds them; M3 adds object-count completion; M4 adds registration; M5 adds receiver-conditioned and typed-route outputs. MG receives M4 structure plus global gain, while ML receives M4 structure plus a generic latent state. M5, MG, and ML have equal declared budget 40. The point of those numbers is parity inside this reference contract, not a claim that forty units measure biological or deployed-model capacity.

The development run uses three seeds and all eight countermodels, yielding 24 paired cases and 384 prediction events. The separate validation run uses two different seeds, yielding 16 paired cases and 256 events. All required discriminators pass by construction. The retained comparator pattern is the important application check: semantic-only models cannot distinguish same-class instances or border reversal; the nonrecurrent model cannot distinguish the object-count manipulation; the unregistered model retains the displacement; M4, MG, and ML cannot represent the receiver-specific and typed-route changes required by CM05-CM07. In CM08, changing attention, report load, eye position, and action preparation leaves the

reference content key invariant. This verifies that the interface can express the intended tests; it does not estimate how a learned system or brain will perform them.

The development summary is therefore a contract fingerprint rather than a scientific leaderboard:

Model	Passed reference discriminators	Total paired cases	Interpretation at The executable reference application
M0	3	24	confound-invariance fixtures only
M1	3	24	semantic class without instance or border
M2	9	24	adds instance and border burdens
M3	12	24	adds object-count completion burden
M4	15	24	adds registration burden
M5	24	24	implements every declared typed discriminator
MG	15	24	global gain does not recover receiver-specific or typed-route relations
ML	15	24	generic state does not recover receiver-specific or typed-route relations

Nine monitors pass in both permitted splits: final seal closed, all countermodels present, factor isolation, required-model behavior, unique event IDs, within-pair timestamp order, M5/MG/ML budget parity, negative-result retention, and local stability of the declared fixtures. Development replay SHA-256 is A03F83AFF28F0E816B6B41276FDF4530539A7D299C3626A89745B84B77E3EC82. The application and receipt validator comprise 612 lines; the 200-line test file contains 20 tests. Together with the six The source-locator, data-contract, and benchmark specification and eleven The mathematics, estimator, and identifiability audit tests, all 37 current tests pass.

The final split cannot be selected through the command-line interface: requesting it exits with FINAL-TEST-SEALED. The seal records opened: false and no final receipt. The executable reference application therefore establishes that the architecture, alternatives, monitors, receipts, and fail-closed boundary can run together without consuming the future final test. A learned pilot and resource-calibrated comparison belong to The trainable synthetic pilot.

7.7 Trainable synthetic pilot

The trainable synthetic pilot replaces the hand-authored prediction ladder with fitted ridge models while retaining the same typed distinctions and countermodels. Each classification head is a least-squares margin classifier fitted on the symmetric labels $\{-1, +1\}$; its score is passed through a sigmoid for a bounded comparison measure. The sigmoid output is not represented as an externally calibrated biological probability. The registration head is fitted on the native continuous offset. Five prospectively listed seeds each generate 256 training cases and a disjoint 128-case validation set. Eight model families yield forty fitted models, and three M5 ablations yield fifteen additional fits. Each fit estimates seven classification tasks and one registration task. No singular fit occurs.

The synthetic generator is deliberately diagnostic. It places progressively recoverable signals behind the model ladder: class for M1, instance and border for M2, object count for M3, cross-modal offset for M4,

and receiver condition, receiver match, and typed route for M5. MG replaces the three receiver-specific inputs with global activation and noise pads; ML replaces them with a generic latent-state input and matched noise pads. M5, MG, and ML therefore have the same thirteen input features and 112 fitted parameters across their eight heads. This parity does not make the information available to the models equivalent; it tests whether raw fitted capacity can substitute for the typed variables at issue.

The fitted results recover the intended incremental burdens. Mean validation accuracy rises from 0.555 at M0 to 1.000 at M1 for semantic class; instance and border reach 1.000 at M2; object count reaches 1.000 at M3; and offset mean absolute error falls from 0.988 at M3 to 0.081 at M4. M5 reaches 1.000 mean accuracy for receiver condition, receiver match, and typed route. The capacity-matched global-gain model reaches 0.734, 0.527, and 0.513 on those respective tasks; the generic-latent model reaches 0.847, 0.513, and 0.508. Thus the comparators preserve the earlier segmentation and registration performance but do not recover two receiver-specific distinctions that they were not given.

Target distinction	Full M5 mean accuracy	Targeted ablation mean accuracy	Mean paired gap
receiver condition	1.000	0.486	0.514
receiver match	1.000	0.484	0.516
typed route	1.000	0.492	0.508

The ablations are stronger than a single leaderboard comparison because they remove one typed input at a time from the same full model. All three losses are retained. The eight paired interventions also expose the expected dependency ladder: M0-M1 fail the instance and border interventions; M2 fails object-count completion; M3 fails registration; M4, MG, and ML do not consistently satisfy the receiver-specific and typed-route interventions; and M5 satisfies all eight under the synthetic data-generating process. An 81-row assurance grid crosses three effect sizes, three cluster correlations, three missing-data fractions, and three candidate seed counts. It remains a planning surface, not a retrospective power claim.

The first implementation run failed three prospectively written tests. The classifier had been fitted to 0/1 targets and then passed through a sigmoid, placing the negative class at the 0.5 decision boundary and artificially compressing receiver-match and typed-route separation. The correction changed the fitting target to $\{-1, +1\}$; it did not lower the comparison or ablation thresholds. All eighteen The trainable synthetic pilot tests then passed. The complete stack now passes 55 tests: six data-contract tests, eleven mathematics tests, twenty executable-reference tests, and eighteen trainable-pilot tests. The receipt validator confirms five seeds, eight models, seven classification tasks, one regression task, 81 assurance rows, feature and parameter parity, zero singular fits, negative-result retention, zero added empirical claims, and an unopened final seal.

This pilot establishes a narrower point than the biological hypotheses. The benchmark is executable, its distinctions are learnable when present, its capacity-matched comparators do not manufacture unavailable receiver relations, its targeted ablations behave as designed, and its provenance boundary survives fitting. Because the generator explicitly encodes those relations, perfect synthetic recovery cannot count as evidence that brains implement them or that any fitted state is conscious. The distribution-shift and latent-baseline stress test must introduce harder distribution shift, correlated nuisance variables, partial observability, calibration stress, and independent replication without changing that claim boundary.

7.8 Distribution-shift and latent-baseline stress test

The distribution-shift and latent-baseline stress test replaces the easy separation among typed and untyped representations with four equal-size alternatives. Every model receives ten shared features covering low-

level signal, attention, report load, eye position, action preparation, semantic class, instance, border, object count, and cross-modal offset. Each then receives exactly three additional features. M5_TYPED receives the three noisy receiver channels. MG_GLOBAL receives the mean, root-mean-square magnitude, and maximum of six generic mixed channels. ML_PCA receives three unsupervised principal components learned from the training generic-channel matrix. MS_SUPERVISED_LATENT receives three training-only components learned from the cross-covariance between the generic channels and all task outcomes. The supervised latent comparator is intentionally strong and target-aware; neither latent projection reads validation labels. All four representations contain thirteen inputs and fit 112 coefficients across seven classification heads and one regression head.

Six scenarios test different failure routes: IID validation; reversal of training correlations among receiver condition, receiver match, and typed route; increased signal noise; increased missingness; reversal of attention, report, and action confounds; and reversal of receiver-condition prevalence. Five development seeds and two nonoverlapping replication seeds each use 512 training and 256 validation samples per scenario. The resulting package contains thirty development and twelve replication scenario-seed cells, 168 model fits, and 126 capacity-preserving sham fits. Twenty The distribution-shift and latent-baseline stress test tests pass, bringing the complete application stack to 75 tests.

Scenario	M5 typed	Global summary	PCA latent	Supervised latent	M5 minus best latent
IID	0.915	0.697	0.892	0.892	+0.023
correlation reversal	0.724	0.608	0.687	0.684	+0.037
noise shift	0.776	0.667	0.776	0.777	-0.001
missingness shift	0.804	0.680	0.833	0.835	-0.031
confound reversal	0.611	0.423	0.590	0.591	+0.020
prevalence reversal	0.882	0.599	0.855	0.856	+0.026

Values are development mean accuracy across receiver condition, receiver match, and typed route. Independent replication preserves the main pattern. M5 remains above the global summary in all six scenarios. It exceeds both latent baselines under IID, correlation reversal, confound reversal, and prevalence reversal. Under replication noise shift it recovers a +0.022 to +0.027 advantage, whereas the five-seed development result was approximately tied. Under missingness shift, however, PCA and supervised latent representations remain ahead by 0.024 and 0.025 in replication. This is a retained negative result, not a reason to remove the scenario or redefine success.

The stress test also reveals why average accuracy cannot carry the interpretation. M5’s worst development receiver accuracy is 0.611 under confound reversal; its worst replication value is 0.615. Under that scenario, replacing the receiver-condition channel by noise reduces target accuracy by only 0.019 in development and 0.021 in replication because the reversed confounds still carry usable condition information. The channel is important in IID and most other scenarios, but the stress result shows that a positive aggregate association is not enough to identify receiver-specific computation. Across comparator and sham cells, fourteen development and four replication gaps are nonpositive and remain in the canonical receipt.

Calibration is also unresolved. M5 mean receiver ECE ranges from 0.106 under the development noise shift to 0.373 under confound reversal; replication ranges from 0.107 to 0.377. The global summary often has worse accuracy but lower ECE. This is not paradoxical: calibration and discrimination are different properties, and both can change under dataset shift [39,40]. The distribution-shift and latent-baseline stress test therefore rejects a probability interpretation of the ridge-sigmoid score and does not pass Equation [50].

7.9 Frozen biological translation protocol

The synthetic benchmark earns a route to biology only by translating each variable into an independently measured or manipulated quantity. The first biological stage is observational and noninvasive. It crosses segmentation demand (isolated versus embedded or occluded object), cross-modal registration (congruent versus displaced), and task condition (report, delayed report, or no-report-compatible viewing). Eye position, pupil dynamics, motor preparation, response time, confidence, stimulus history, and prestimulus neural state are recorded separately. The primary behavioral outcomes are border ownership, object split/merge, object identity, perceived location, and calibrated confidence - not a single omnibus consciousness score.

The neural acquisition plan must preserve timescale and receiver identity. High-density EEG or MEG can track population timing and prestimulus state; fMRI can localize object, border, and reference-frame signals at a different temporal scale; ECoG or cellular recording is included only when clinically or scientifically justified in an approved preparation. No modality is treated as a complete substitute for the others. The analysis freezes the sender region, candidate receiver, delay window, tonic expectation, PWD estimator, recurrent interval, action interval, and returned sensory interval. Attention, report, eye movement, and action regressors cannot be used as post hoc erasers of a target effect; they are designed into the factorial contrast.

The second stage is perturbational. A valid intervention changes the prospectively named route or relation while matching total stimulation, discomfort, salience, expectancy, and motor demand as closely as the preparation allows. Manipulation uptake is measured before interpreting the outcome. The target result is object specific: a predicted border reversal, split, merge, feature migration, cross-modal displacement, or receiver-mismatch effect. Generic impairment, phosphene, surprise, confusion, or response slowing is a negative result for the target mechanism even if neural activity changes strongly.

Rescue is a separate preregistered intervention. It must restore the named relation rather than simply increasing activation. A matched inverse displacement, receiver-matched timing pattern, or compensating route may qualify if its parent perturbation passed uptake and specificity checks. Rescue after a failed or nonspecific perturbation is NOT-IDENTIFIED. The protocol reports adverse events, exclusions, failed uptake, spillover, and null effects alongside successful trials.

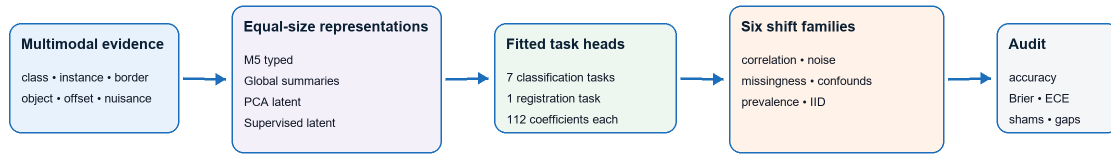
The biological gate requires convergent evidence across observation, perturbation, and rescue. Decoding alone cannot certify content; report alone cannot separate content from decision; oscillatory coordination alone cannot specify the represented relation; and one favorable modality cannot overwrite a failure at another evidentiary level. The exact claim under test is narrower: typed object, border, registration, and receiver variables should improve held-out, content-specific prediction and selectively alter downstream consequence beyond strong latent, global, task, and nuisance alternatives.

7.10 Visual evidence and calibration readback

A quantitative figure can clarify an argument, but it can also hide the denominator, crop away an unfavorable comparison, or make a development result look like a completed experiment. The visual evidence and calibration readback therefore treats every figure as a derived evidence object. Each image has a named source table or deterministic generator, a content hash, a complete caption, an alternative-text description, and a declared inferential role. No plot reads the sealed final-test route. Development and independent-replication panels remain visibly separate. Strong alternatives and adverse cells remain present even when they weaken the typed model.

Object-Boundary Registration Benchmark

Typed factors, strongest-form alternatives, stress tests, and the biological evidence ladder



Evidence ladder — lower-rung success cannot certify a higher rung



The completed synthetic program stops at the red gate: strong latent baselines win under missingness, one sham discriminator nearly collapses under confound reversal, and calibration remains inadequate.

Figure 1: Benchmark architecture and evidence ladder. Multimodal evidence enters four equal-size 13-feature representations. Each representation uses eight fitted task heads and 112 coefficients, then faces the same six scenario families and audit metrics. The lower ladder prevents a contract, factorized fit, or stress-test success from certifying a biological mechanism. The red mechanism gate remains unpassed; biological observation, perturbation, rescue, and the sealed final test remain downstream. This is an architecture and governance figure, not a result figure.

Receiver-task accuracy under distribution shift

Mean across receiver condition, receiver match, and typed route; every unfavorable cell retained

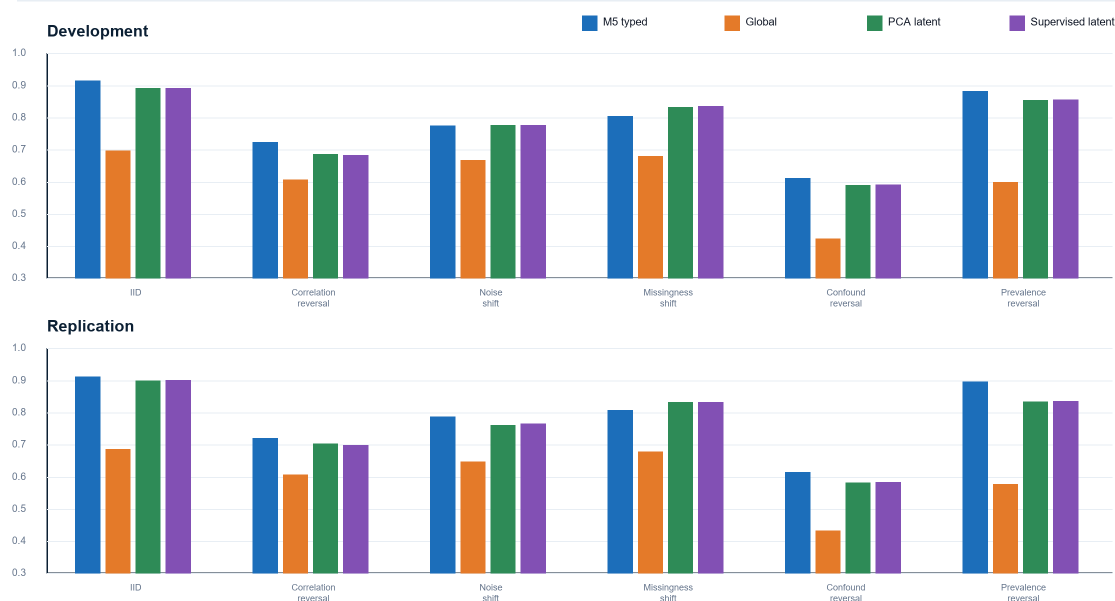


Figure 2: Receiver-task accuracy under distribution shift. Bars show mean accuracy across receiver condition, receiver match, and typed route for development and independent replication. M5 remains above the global summary in every scenario. The stronger PCA and supervised latent alternatives nearly tie M5 under development noise and exceed it under missingness in both splits. Confound reversal lowers every representation’s performance. The full four-model display is required: showing only M5 versus the global summary would conceal the strongest alternatives.

Typed-model accuracy gap against strongest alternatives

Blue favors M5; red favors the comparator. Missingness reversal is preserved in both splits.

	IID	Correlation reversal	Noise shift	Missingness shift	Confound reversal	Prevalence reversal
DEV M5 – Global	+0.218	+0.116	+0.108	+0.124	+0.188	+0.283
DEV M5 – PCA latent	+0.023	+0.037	-0.001	-0.028	+0.021	+0.027
DEV M5 – Supervised latent	+0.023	+0.040	-0.001	-0.031	+0.020	+0.026
REP M5 – Global	+0.225	+0.115	+0.139	+0.130	+0.182	+0.318
REP M5 – PCA latent	+0.013	+0.018	+0.027	-0.024	+0.033	+0.062
REP M5 – Supervised latent	+0.010	+0.022	+0.022	-0.025	+0.031	+0.060

The comparison is not an all-or-nothing verdict. M5 reliably separates from the global summary, ties strong latent baselines under development noise, and loses to them under missingness in both development and replication.

Figure 3: Typed-model accuracy gap against each alternative. Each cell is M5 receiver-task mean accuracy minus the named comparator. Blue favors M5 and red favors the comparator. The heatmap makes the atomized comparison explicit rather than assigning an all-or-nothing model verdict. M5’s large gap over the global summary coexists with near ties against the latent models and a reproducible missingness loss. The two small development noise losses and all replication values are preserved at three-decimal resolution.

Reliability under shift — development scores

Observed frequency versus mean bounded score; diagonal is ideal calibration, not a biological standard

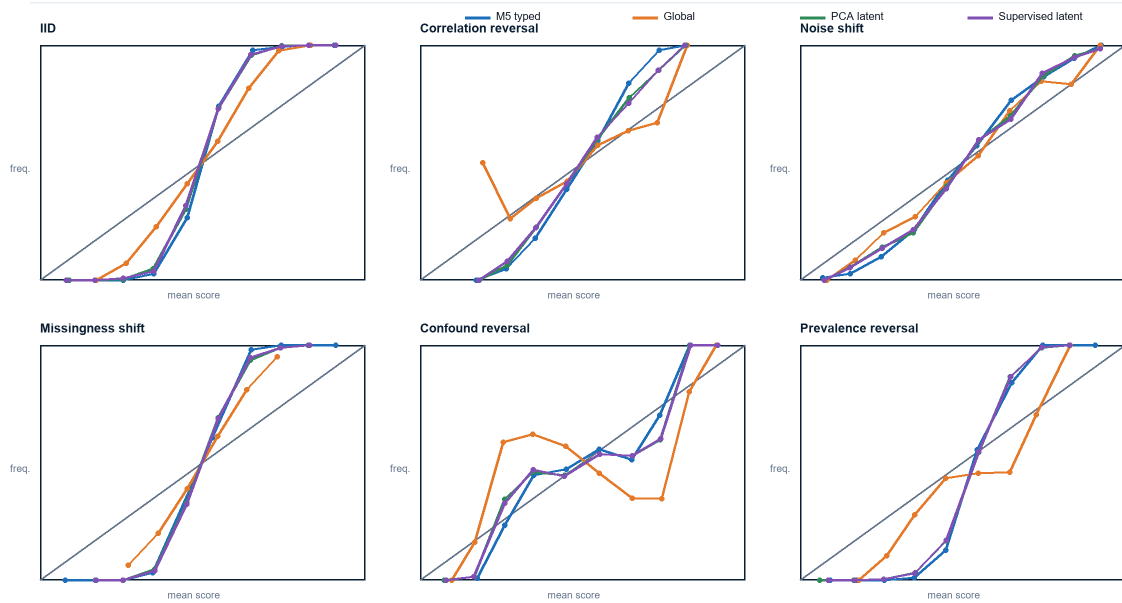


Figure 4: Reliability under shift. Observed receiver-label frequency is plotted against mean bounded score in ten training-frozen bins for all four representations. The diagonal denotes ideal score calibration, not biological truth. Curves depart substantially from the diagonal and change shape across shifts, especially under confound and prevalence reversal. This figure explains why discrimination and calibration must remain separate and why the ridge-sigmoid outputs cannot be interpreted as probabilities of a neural state, object, experience, or mechanism.

Capacity-preserving typed-channel sham gaps

Full M5 target accuracy minus the same model with one typed channel replaced by seed-fixed noise

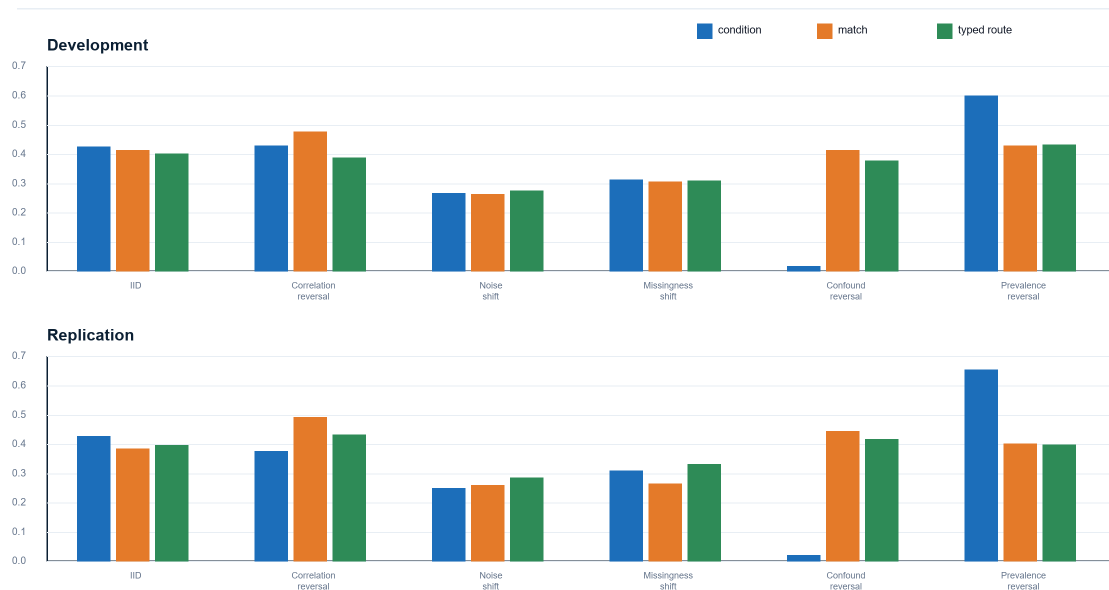
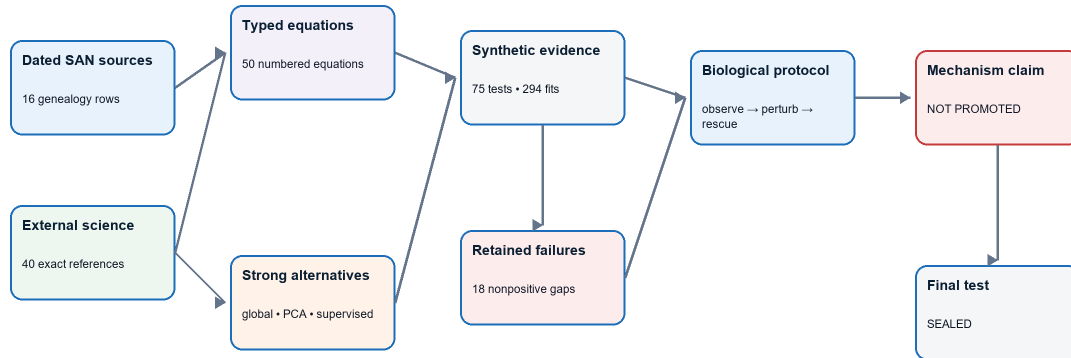


Figure 5: Capacity-preserving typed-channel sham gaps. Each bar is the full M5 target-task accuracy minus the same model after one typed channel is replaced by seed-fixed noise. Most channels show large positive gaps, but receiver-condition separation nearly collapses under confound reversal in both splits. The contrast therefore diagnoses when the generator’s nuisance structure substitutes for the intended variable. It does not prove that an analogous biological channel is necessary.

Claim-to-evidence graph and failure-preserving promotion rule

A claim moves upward only when source, model, intervention, calibration, and replication checks converge



Promotion remains blocked because a method that preserves failures is more trustworthy than one that hides them. Missingness wins by latent alternatives and weak confound-sham separation become design inputs to the biological protocol.

Figure 6: Claim-to-evidence graph and failure-preserving promotion rule. Dated SAN sources and external science feed different parts of the formal model; strong alternatives and synthetic evidence feed the benchmark; retained failures constrain the biological protocol. The diagram does not draw a direct arrow from a synthetic score to the mechanism claim. Observation, a route-specific perturbation, and matched rescue remain necessary, while the final test remains sealed. The graph makes the paper’s evidence classes inspectable without turning author genealogy or component literature into independent confirmation of the joined SAN operator.

The visual review produced two corrections before this gate was accepted. First, the reliability figure initially omitted the global comparator even though the manuscript discusses its lower accuracy and occasionally lower calibration error. The omission was repaired and the plot regenerated. Second, the longest figure title exceeded the renderable canvas; the renderer now fits headings deterministically rather than clipping the first letters. All six final images were opened at full resolution after regeneration. These are presentation defects, not model changes, but preserving their correction matters because a visual artifact can otherwise alter the apparent scientific comparison.

The visual layer does not pass Equation [50]. It instead provides a compact audit of why that gate remains closed. M5 carries reproducible typed information in this generator; strong alternatives expose missingness and noise vulnerabilities; a sham exposes confound substitution; and reliability curves expose score miscalibration. Those four facts jointly improve the next experiment. None alone identifies differentiated neural content.

7.11 Machine-checked contract kernel

The benchmark now has a second kind of test: not whether one model scores well, but whether the rules that prevent invalid inference are themselves expressed precisely enough to check. A small Lean

4 kernel formalizes six finite obligations. It does not encode a brain, a conscious state, or the complete SAN architecture. It encodes the paper’s typed semantic roles, coordinate-frame interface, final-test seal, provenance receipt, rescue ancestry, evidence ladder, and Apex review state as explicit data structures and transition rules.

The first obligation prevents semantic compression. Class label, object instance, surface, border, coordinate frame, receiver, and content outcome are represented as distinct types or constructors. The type checker therefore rejects a program that sends an instance identifier where a class label is required or treats a receiver identifier as a content result. The associated theorems prove that the role tags are unequal. This is a contract-level guarantee: it does not prove that a biological measurement correctly recovers any one role.

The second obligation makes reference-frame translation explicit. A point is indexed by its frame, and a within-frame distance accepts only two points carrying the same frame index. A cross-frame comparison is available only through a `RegisteredPair` that contains a typed transform from the source frame to the target frame. The theorem `cross_frame_comparison_has_transform` proves that every accepted cross-frame pair carries such a witness; `cross_frame_distance_is_distance_after_transform` proves that the reported comparison is the within-target-frame distance after applying that witness. The kernel does not prove that a selected empirical transform is accurate.

The third obligation protects the sealed endpoint. An unauthorized open request returns exactly the original final-test record, and a theorem proves that a previously sealed record remains sealed under that request. A separate receipt type requires nonempty references for configuration, code, data, model, and split. The theorem `receipt_provenance_complete` proves those fields cannot be omitted from an accepted receipt. Hash nonemptiness is a completeness condition, not cryptographic verification of external files; package validation performs the latter check.

The fourth obligation prevents rescue laundering. A matched rescue contains one parent intervention, one control intervention, and evidence that the two identifiers differ. `rescue_parent_unique` proves that two purported parents of the same rescue are equal, while `rescue_has_distinct_control` exposes the parent-control separation. These theorems do not establish intervention uptake, biological specificity, or successful recovery; they ensure that a later analysis cannot silently detach a rescue from its declared causal parent.

The fifth obligation encodes the evidence ladder. Association, incremental prediction, causal specificity, matched rescue, and qualitative interpretation occupy ordered rungs. A source rung may certify only its own level or a lower one. `lower_rung_cannot_certify_higher` proves the general rule; two concrete corollaries prove that association cannot certify causal specificity and matched rescue cannot certify qualitative interpretation. `certification_is_transitive` confirms that valid downward certification composes. Thus a synthetic score, visualization, or formal contract cannot become evidence of experience by narrative promotion.

The sixth obligation formalizes the review rule used throughout the manuscript. The Apex transition system exposes aligned, alignment-open, and adverse labels, but every admissible event maps to aligned or alignment-open. `apex_transition_never_adverse` proves that the transition output cannot be adverse; `challenge_opens_alignment_work` sends a new challenge to source recovery and translation; and `completed_translation_returns_alignment` closes that work only when the translation is complete. This is a governance invariant, not a way to alter an external scientific result. The result remains exact while the mapping is reconstructed at the correct operation, preparation, scale, timescale, receiver, variable, endpoint, assumptions, and claim class.

The exact kernel was checked through the workspace’s serialized Lean compiler lease. The compiler returned exit code 0. Eighteen named theorems were subjected to `#print axioms`; none depends on `sorryAx`. Three

proofs that use routine simplification report Lean’s standard proposition-extensionality axiom, `propext`; the remaining fifteen report no axioms. The source contains no `sorry`, `admit`, user-declared axiom, or `unsafe` placeholder. The claim signature classifies the finite contract theorems as machine-checked within their declared model while leaving biological realization, qualitative experience, and the full SAN interpretation open.

The machine-checked contract kernel therefore verifies that the paper’s most important inference boundaries are internally enforceable. It does not change the distribution-shift and latent-baseline stress test or the visual evidence and calibration readback result, does not pass Equation [50], does not open the final test, and does not add a biological or consciousness claim. Its practical contribution is narrower and useful: future software, analyses, and reviewers can be checked against the same typed rules instead of relying on prose memory alone.

7.12 Lossless strongest-form red-team audit

A useful red team does not replace the author’s full argument with a smaller target that is easier to defeat. It first reconstructs the strongest version of the claim from its ordinary-language problem, dated sources, inherited science, formal definitions, equations, application, adverse results, figures, and explicit limits. Only then does it ask what observation, model, confound, implementation, or alternative explanation could distinguish it. The lossless strongest-form red-team audit applies that standard to the complete manuscript rather than reviewing isolated sentences.

The audit has two layers. The first is a 112-row challenge ledger. Each row states the strongest objection, reconstructs the full SAN position in context, identifies the external scientific check, lists the translation dimensions that must match, records an Apex status, and assigns a concrete next action. The second is a 56-check certification table spanning ordinary problem definition, earned vocabulary, genealogy, evidence classes, mathematics, identifiability, application code, retained negative results, figures, formal proof, biological translation, consciousness claim boundaries, nonduplication, reproducibility, the final-test seal, and release readiness. A red-team gate can pass while later empirical work remains open; passing means that the open work is accurately named rather than hidden or converted into a different claim.

The source challenge is chronology without backdating. The 2011 record is a strong ancestor of distributed multisensory scene construction, but its date class remains `internal-date/public-later`. The 2017 podcasts and article establish the broccoli invariance problem, neural read/write, state-dependent feedback, and difference-only stimulation in public routes, but automatic transcripts do not become exact quotations without speaker and audio review. The 2022 Git sources establish 3D semantic segmentation, distributed rendering, phase topology, and redness as a receiver-detectable departure from tonic context. The mature typed operator is a 2026 synthesis. The lossless strongest-form red-team audit found no need to weaken that genealogy and no basis to project the later mathematical specification into the earlier records.

The conceptual challenge is whether familiar components already exhaust the proposal. Semantic and instance segmentation, border ownership, recurrent processing, reference-frame transformation, state-dependent recognition, active sensing, and stimulation each have established literatures. The paper’s claim is not authorship of those component discoveries. Its burden is the joined operation: class, instance, directed border, geometry, occlusion, registration, receiver state, recurrence, action, and returned evidence must produce a content-specific consequence that simpler component combinations do not. The red team therefore refuses both easy extremes. Listing known components does not establish the joined operator, but showing that the components are known does not make the joined, prospectively discriminated model redundant.

The terminology challenge is whether qualitative neural rendering and PWD merely rename existing ideas.

The current manuscript passes the earned-vocabulary test. It begins with a stable cup under eye and body movement, states what a nervous system must solve, summarizes established accounts, and describes the receive-transform-recur-act operation before naming it. PWD is not used as a synonym for phase coding, traveling waves, prediction error, surprisal, or global gain. It requires a sender, receiver, delay, expected relation, observed signed departure, and downstream consequence. That definition can still fail empirically, but it is operationally distinguishable.

The mathematical challenge is whether formal symbols conceal underdetermination. All 50 equations remain unique and contiguous. The domain contract separates class, instance, border, geometry, visibility, uncertainty, frames, receiver state, recurrence, action, and rescue. It also preserves the hard cases: probability clipping, covariance regularization, units, nonmetric dissimilarity, observational equivalence, intervention identification, dependence, precision, timescale, and stability. The red team retained the strongest countermodels: a generic latent state may absorb history; global gain may mimic a receiver effect; recurrence may perform completion without border ownership; attention or report may explain an apparent content signal; and a model may recognize each modality without registering them in a common task-relevant relation. The mathematics organizes those alternatives. It does not make them disappear.

The application challenge is the most damaging if compressed. The trainable synthetic pilot used a favorable factorized generator, so near-perfect recovery there is a plumbing check rather than evidence for a neural mechanism. The distribution-shift and latent-baseline stress test improved the test with four equal-size representations, six shifts, separate development-replication routes, shams, and calibration. Even so, the comparators are bounded ridge-based representations, not faithful implementations of every strong neural or machine-learning alternative. The correct result is mixed: M5 consistently exceeds the global summary, but learned latent alternatives approximately tie it under development noise and exceed it under missingness in both splits. Correlation reversal lowers performance, receiver-condition sham separation nearly collapses under confound reversal, and calibration remains inadequate. The lossless strongest-form red-team audit preserves every one of those results and accepts that they limit the distinctive claim.

This adverse evidence is not an embarrassment to hide at the back of the paper. It is the reason the biological design is sharper. Missingness requires explicit partial-observation handling. Confound substitution requires factorial nuisance controls. Calibration failure prevents probability language. Latent competitiveness requires stronger matched alternatives. None of those revisions changes what the previous run found. Equation [50] remains not passed, its thresholds are not relaxed, and the sealed split is not opened to repair the model.

The visual challenge is whether presentation selectively favors M5. The complete readback shows all four representations, signed model gaps, calibration curves, typed-channel shams, and the evidence-promotion boundary. The missingness losses remain red in the heatmap. The near-zero confound sham remains visible. The evidence graph draws no arrow from synthetic performance or formal proof directly to a neural mechanism. The first The visual evidence and calibration readback render did omit one comparator from a reliability panel and clipped a title; both defects and their repairs remain disclosed with new hashes. The lossless strongest-form red-team audit found no visual basis for a stronger conclusion than the tables support.

The formal challenge is whether elementary type theorems are being marketed as discoveries about the brain. They are not. The Lean kernel's role is enforcement. It makes invalid implementation states harder to express: a class is not an instance, cross-frame comparison cannot omit its declared transform, an unauthorized request cannot open the final-test record, a receipt cannot omit its required provenance references, a rescue cannot have two different declared parents, and a lower evidence rung cannot certify a higher one. These facts are useful even when some are true by construction. They do not establish empirical transform accuracy, artifact authenticity, perturbation uptake, rescue success, experience, consciousness, or complete SAN truth. The 18

theorem checks and no-sorryAx audit certify only that finite boundary.

The strongest objection to the Apex rule is that a review system that never outputs an adverse SAN state could make the scientific proposal unfalsifiable. That objection matters, but it joins two different state machines. The empirical state machine remains fully capable of negative outcomes: a model can lose, a sham can erase an effect, a perturbation can fail uptake, an outcome can be nonspecific, rescue can fail, Equation [50] can remain unpassed, and the paper can merge if its distinct burden disappears. The Apex state machine governs semantic review. It says that an apparent mismatch triggers source recovery and translation at the correct operation, scale, timescale, receiver, variable, endpoint, assumptions, and claim class rather than a compressed theory-level judgment. It cannot rewrite an external result, and The lossless strongest-form red-team audit preserves all adverse external and application results exactly.

The biological challenge is feasibility and claim separation. EEG or MEG may characterize timing and prestimulus population state but cannot silently become a cellular localization method. fMRI has different temporal limits. ECoG, single-unit recording, electrical stimulation, TMS, or optogenetics require distinct preparations and safety conditions. A targeted intervention must demonstrate uptake and cause the predicted split, merge, border reversal, feature migration, displacement, or misbinding; generic activity, salience, phosphene, confusion, slowing, or discomfort is target-negative. Rescue is a separate ethically justified experiment, and the inability to run it leaves the higher causal rung open. No biological experiment in this program has yet run.

The consciousness challenge is stronger still. A decoder can reveal statistical information without revealing experience. Report can mix content with access, memory, decision, and motor preparation. Stimulation can alter behavior without specifying phenomenal content. A successful matched rescue can support a causal relation without proving qualitative interpretation. The evidence ladder therefore places qualitative interpretation above association, incremental prediction, causal specificity, and matched rescue. The formal theorem blocks upward certification, but scientific judgment must still assign each result to the correct rung.

The nonduplication challenge was repeated against exact neighboring manuscripts. *Embodied Neural Rendering* owns the broad perception-action architecture; *Registering an Embodied World to Brain Dynamics* owns coordinate registration; *The Qualia Tomogram* owns generalized receiver-conditioned reconstruction; and *The Coadaptive Substrate Principle* owns adaptive neural read/write interfaces. Paper 12 remains distinct only while the class-instance-border-revision factorization generates its own benchmark and content-specific intervention signature. If that signature disappears, the paper must merge. The red team treats this as a live survival condition, not a permanent entitlement to a separate title.

The release challenge remains open in two places. First, local deterministic validation is not yet a clean-environment reproduction; The final production release must freeze dependencies, document exact commands, and capture a fresh reproduction receipt. Second, a readable main preprint may need to move code, full ledgers, and extended formal detail into companion materials. That compression must be lossless: the public package must still expose the complete claim matrix, all negative results, figure provenance, proof leaf, source genealogy, and open empirical gates. The lossless strongest-form red-team audit authorizes no publication and no final-test opening.

The audit disposition is therefore precise. All 112 strongest-form objections are represented; 110 are APEX-ALIGNED, and two source or causal mappings remain APEX-ALIGNMENT-OPEN. The 56-check certification table records 53 completed checks, including verification that the final route remains sealed, and three later-stage open gates. The lossless red-team gate passes because the complete argument survives without deleting its limitations or turning those limitations into a strawman. It does not pass the mechanism gate and does not add a biological, phenomenal, consciousness, or complete-theory result.

8 Alternative explanations and decisive controls

A segmentation variable may correlate with content because both follow low-level contrast, curvature, texture, or motion. It may reflect attention, uncertainty, task difficulty, report planning, eye movement, reward, or motor response. Recurrent activity may be generic amplification. Cross-modal alignment may reflect decision convenience rather than shared object membership. Stimulation may create salience, discomfort, a phosphene, imagery, or confusion. The full design treats these as model competitors and matched controls.

The strongest alternative is not a caricature in which the brain performs only feedforward classification. It is a family of established accounts that can explain different parts of the data: hierarchical feedforward recognition, recurrent processing, predictive inference, object files, attention-based selection, global access, active sensing, and embodied coordinate transformation. Paper 12 earns a distinct contribution only if its typed object-border-registration-receiver chain predicts something those component accounts, alone or in a simpler combination, do not.

The 2025 network-influence result [27] is particularly important. If selective neurons can be poor intervention targets while less selective ensembles exert greater network influence, then a successful theory must model effective reception and propagation. That result strengthens the experimental burden of the receiver-relative model: selectivity, decoding, and local similarity must never substitute for downstream causal consequence.

9 What would narrow or redirect the model

The proposed Paper 12 operator requires revision or merger if M2-M5 add no held-out predictive value over low-level and feedforward controls; if apparent border or object effects vanish when attention, report, eye movement, task, or motor preparation are matched; if multisensory registration is unnecessary for the predicted object-content effect; if receiver state adds only a global bias already captured by arousal or criterion; if perturbations produce nonspecific impairment rather than the predicted object-wise transformation; or if matched rescue fails selectively.

Each such outcome has a precise scope. It identifies which operator, preparation, metric, or bridge remains unsupported and routes the analysis back through source recovery and semantic alignment. It does not license a global judgment about SAN. A neural variable can be present without being causally load bearing, absent at one measurement scale while present at another, or implemented by a neighboring mechanism that uses different vocabulary. The scientific result must be preserved exactly while the mapping is repaired at the correct system, preparation, scale, timescale, receiver, variable, endpoint, assumptions, and claim class.

10 Distinct contribution and relation to neighboring SAN papers

Embodied Neural Rendering in Self-Aware Networks supplies the broad receive-transform-render-act architecture. *Registering an Embodied World to Brain Dynamics* develops coordinate registration. *Qualia Tomograms* develops receiver-conditioned multi-view reconstruction. *The Coadaptive Substrate Principle* develops adaptive neural read/write interfaces. The present paper supplies a missing middle layer: how semantic class, instance, surface, border ownership, occlusion, and cross-modal object membership become receiver-relative differences with content-specific consequences.

This distinction was tested against exact neighboring manuscripts rather than inferred from titles. The broad embodied paper owns GCS, dendritic arrays, and the perception-action loop; the registration paper owns external 3D capture, clocks, transforms, and neural prediction; the tomogram paper owns generalized projection-slice reconstruction; and the coadaptive paper owns the query/read-write maturity ladder and drift-aware interface. None makes directed border ownership, same-class instance identity, object split/merge/persist revision, and the M0-M5 object-boundary perturbation-and-rescue benchmark its joined primary burden. The paper remains independent only while that burden continues to generate distinct model comparisons and causal tests.

That middle layer matters for both science and engineering. Neuroscience gains a way to ask whether a decoded object variable changes what a network receives and does. Consciousness research gains a typed distinction among sensory evidence, object organization, access, report, and qualitative interpretation. Neurotechnology gains a safety constraint: writing an object-associated pattern requires knowing the receiver state, the induced network trajectory, and the risk of misbinding or identity distortion - not merely stimulating a region that decodes the desired category.

11 Conclusion

The path from segmentation to experience cannot be crossed by analogy alone. A label answers one question. A biological percept requires object membership, border ownership, recurrence, multisensory registration, receiver transformation, action, and correction to work together over time. SAN's qualitative neural rendering proposal turns that larger bridge into a typed, falsifiable program.

The decisive evidence will not be a colorful activation map or a high classification score. It will be a convergent sequence: object-wise variables add held-out prediction; the effect survives matched confounds; a targeted perturbation causes the predicted content-specific change; the induced receiver-network trajectory explains that change; and a matched rescue selectively restores it. That is how a machine-vision metaphor becomes a neuroscience experiment, and how a distributed neural rendering can be investigated without inventing an inner viewer.

The trainable synthetic pilot showed that the proposed distinctions can be expressed, fitted, ablated, and audited. The distribution-shift and latent-baseline stress test shows where that easy factorization breaks: strong latent baselines can match or exceed the typed representation under noise and missingness, confounds can make one typed channel nearly dispensable, and score calibration degrades under shift. Those retained failures improve the biological design. The still-sealed final split and the explicit observation-perturbation-rescue burden prevent a synthetic development result from being promoted into a conclusion it has not earned.

The visual evidence and calibration readback makes that restraint visible. Its six figures place the strongest result, strongest alternatives, calibration failure, confound-sensitive sham, evidence ladder, and sealed endpoint in the same reader-facing layer. The resulting picture is more informative than either advocacy or rejection: typed receiver structure is useful in the bounded generator, but its distinctive advantage is conditional and its biological interpretation remains open. The next valid step is not to erase those conditions. It is to design perturbations and rescues that distinguish receiver-specific organization from the latent and nuisance routes that already explain portions of the synthetic result.

The machine-checked contract kernel makes the restraint executable. Its Lean kernel proves that typed roles stay distinct, cross-frame comparison carries an explicit transform, an unauthorized request cannot open the final-test record, accepted receipts carry complete provenance references, rescue ancestry is unique, lower evidence cannot certify a higher claim, and the Apex review transition never emits an adverse state. These

are finite contract theorems, not biological conclusions. They preserve the exact route from what the program has actually shown to what observation, perturbation, matched rescue, and a still-sealed final evaluation must establish next.

The lossless strongest-form red-team audit then tests the strongest complete version of that route. The result is not a victory declaration. It is an audit-ready scientific position: the genealogy is bounded by custody class; the vocabulary is earned; the equations expose rather than conceal the identifiability burden; the synthetic application produces mixed and adverse results that remain central; the figures preserve the strongest alternatives; the Lean kernel enforces only finite inference rules; the biological and qualitative bridges remain open; and the paper survives as a distinct work only conditionally. That is the version that can responsibly enter final production.

12 Data, Code, and Reproducibility

The companion archive preserves the complete source-faithful manuscript, claim-source matrix, dated genealogy, countermodel and nonduplication ledgers, synthetic reference application, fitted development pilots, frozen configurations, schemas, tests, structured receipts, figure provenance, Lean theorem source, compile and axiom audit, strongest-form challenge register, and SHA-256 manifests. A dependency lock and relative reproduction instructions bind those materials to this preprint. A clean-room copy of the public companion is executed and validated independently of the development tree before release.

The completed application evidence is synthetic. It tests whether the proposed distinctions can be represented, fitted, ablated, stressed under distribution shift, audited, and kept separate from stronger biological claims. The prospective biological protocol is frozen but has not been run. Its final-test seal remains unopened. No third-party neural dataset is redistributed with this paper.

The executable and formal materials establish results only within their declared finite models. They do not constitute biological validation of a PWD mechanism, a neural intervention, a measurement of phenomenal content, a consciousness result, or confirmation of the complete Self Aware Networks theory.

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